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Title

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Abstract

We present a randomized algorithm for selecting the median of an unsorted array in $\mathcal{O}(n)$ time with high probability. Unlike classical deterministic methods—notably the median-of-medians algorithm—our approach leverages random sampling to simplify the pivot-selection process and reduce overhead. Specifically, we extract a sublinear sample of the input, sort this sample, and employ its near-median elements as pivots. By bounding $\mathbb{P}[|X - \mathbb{E}[X]| \geq k]$ for appropriate choices of X and k, we ensure that the size of the sub-array requiring a full sort remains sufficiently small. A rigorous analysis, based on Chebyshev's inequality, demonstrates that the algorithm's probability of "failure" (i.e., incorrectly bracketed pivots) decays quickly with the input size. In successful executions, the asymptotic cost of sorting the sample and the sub-array is dominated by a single pass through the input, yielding an overall running time of $\mathcal{O}(n)$. While deterministic linear-time solutions to selection exist, our randomized method offers competitive performance in practice, owing to its simplicity and favorable hidden constants. We also discuss how this technique can be combined with deterministic strategies to eliminate worst-case risk.

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Chapter 1

Justification (temp title)

Core features:

- Granular reentrancy protection
- Side effect tracking
- Algebraic data types
- Safety by default
- Traits instead of inheritance

1.1 Algebraic Data Types

When modelling a state machine, it is standard practice to make use of regular enumeration types. However, such a value can only encode a single state, but not contain any auxiliary information. Standard workarounds include using "magic values", i.e., encoding state information via setting a field to a specific predetermined value, as well as defining a "kitchen-sink" struct, namely one that contains all fields for all states at the same time.

For example, ERC20 tokens are specified such that the value `u256::MAX` $= 2^{256} - 1$ represents an unlimited allowance, whereas any other value represents an allowance equivalent to its actual quantity. Now, any read or write to such an allowance must be checked against this magic value, leading to a higher risk of developer mistakes.

Instead, the allowance could be represented by an algebraic data type with two states, one which represents a specific finite amount and a second which represents the unlimited case. Then, the type system can enforce that the two states can never be mistaken for one another. Any time a value is read, the developer must handle both cases to satisfy the type checker. When a value is written, the developer must specify which state they intend to store.

1.1.1 Real-world example

Smart contracts often are used to model processes that function like state machines, for example voting processes. The widely-used OpenZeppelin contract library features a 'Governor' contract, which models such functionality. Users can submit proposals, each one of which can be in one of eight different states. Depending on the state, a proposal needs to keep track of different values. For instance, while voting is active, it needs to store the deadline for the end of the voting period, as well as the actual vote counts. This is done using the following struct:

```
struct ProposalCore {
    uint48 voteStart;
    uint32 voteDuration;
    bool executed;
    bool canceled;
    uint48 etaSeconds;
}
```

Precisely which state the proposal is in is dictated by the values stored in the fields. For instance, if `executed` is set to `true`, then the proposal is in its executed state and all the other fields are irrelevant. However, if `executed` and `canceled` are both set to `false`, then it depends on whether the current time is before `voteStart`, before `voteStart + voteDuration`, or after both. Thus, even just determining what state the proposal is in becomes exceedingly complex. Instead, it would be much simpler to describe a proposal starting from its states, and including the relevant fields for each state. This is precisely what can be done with an ADT, as well as letting the compiler reject reading field values that are not relevant to the current state. For clarity's sake, here is what such an ADT might look like, in Rust syntax:

```
enum ProposalCore {
    Pending {
        voteStart: u64,
    },
    Active {
        voteStart: u64,
        voteDuration: u64,
    },
    Canceled,
    Defeated,
    Succeeded,
    Queued {
        etaSeconds: u64,
    },
    Expired,
```

```
    Executed,
}
```

This additionally demonstrates well that most fields are irrelevant for most states.

1.2 Reentrancy Protection

Reentrancy vulnerabilities are one of the most widely-known classes of smart contract bugs. Despite the fact that most smart contract developers are aware of their existence, reentrancies are to this day one of the most common vectors for real-world exploits. Thus, it is clear that the tools provided by the current development ecosystem are insufficient to systematically prevent this type of vulnerability.

In fact, Solidity provides no built-in reentrancy protection tools at all, although its *function modifier* feature provides a convenient way for developers to implement their own protection mechanism. Vyper on the other hand does provide opt-in reentrancy protection, although even this protection was exploited due to a compiler bug.
Note that in both cases, the reentrancy checks are specific to their entrypoints. This means that if an entrypoint is not protected, any subsequent external call made by the contract could potentially reenter any of the contract's entrypoints. If an entrypoint *is* protected, any external call made is still able to reenter all unprotected entrypoints of the contract.

It is trivial for the compiler to infer what kind of check needs to be made based on the entrypoint, as well as only allowing specific external calls to reenter only their desired entrypoints.

Only a minority of smart contracts even require reentrant calls for their core functionality, and even fewer of these require anything more than a single reentrant call path. As such, it seems prudent for most cases to disable reentrant calls by default, and allow the developer to individually define each required reentrant call path individually. If the core functionality of a smart contract must include reentrancies, the developer must consider the safety of such calls very carefully. Hence, it is little additional overhead to require them to specify each call path individually. In fact, it requires every necessary call path to be explicitly considered, so it is virtually impossible for an unexpected reentrant call to happen.

1.2.1 Real-world examples

There is an abundance of real-world reentrancy vulnerabilities, although many are either outdated or part of large projects and hence difficult to summarize succinctly.

In October 2022, a new class of reentrancy vulnerability was first described – named a read-only reentrancy. It was long the consensus that reentrancy bugs could only occur if the reentrant call itself modified the contract state. However, simply the fact that a read-only function might return an incorrect intermediate value can be enough to trigger a bug. Most cases of such bugs involve price oracles – that is, contracts whose purpose is to estimate the current exchange price between two or more currencies. It is often possible to manipulate these oracles, although it is usually at great expense. In some cases, it is possible to manipulate the oracle price, then for the oracle to call back to the user. If a third-party contract now relies on the value returned by the oracle contract, it will read the manipulated price and be vulnerable to exploitation. Note that this would not be possible if the oracle contract didn't allow control flow to reenter the read-only function.

In fact, the standard approach to reentrancy protection in Solidity does not even allow protecting read-only functions. Instead, developers must implement two separate checks, one for regular calls and one for read-only calls. Meanwhile, Vyper has extended its opt-in mechanisms to also include read-only functions as well.

todo: reentrancy examples, compound ERC777

1.3 Side Effects

Note: "Side effect" is a very broad term that can be applied to many types of behaviour. In this case, we specifically discuss *state modifications*, but the approach can be extended to almost any other kind of side effect, such as state reads, event emission, external calls, and so on.
In a sense, one could say that the purpose of smart contracts is to cause side effects on the blockchain. In any case, it is quite important for developers to be aware of what side effects are happening in what order. In the first case, hidden or implicit side effects in the code can trigger unexpected behaviour, for example if one state change overwrites another, or if outdated values are used for further computations. In the second case, ambiguities in execution order can also lead to unexpected behaviour, although this is more limited to *intentionally misleading* code, such as the winning submission in the **Underhanded Solidity Contest** in 2022.

The evaluation order of expressions is explicitly stated to be undefined in Solidity, which means that a vast amount of deployed smart contracts are actually inducing undefined behaviour during the compilation process. Of course, this is almost always benign, but still represents a real threat. The Vyper documentation also explicitly states that the evaluation order of arguments to built-in functions is undefined. In other cases, the documentation does not specify anything at all.

Keeping track of state modifications throughout the various call paths of a smart contract is also critical in the case of smart contract audits. If it were at a glance clear that an entrypoint cannot trigger certain state modifications, it would be much easier to rule out certain kinds of vulnerabilities, without having to manually go through each line of code. Additionally, if the syntax is explicitly clear about when side effects can happen, it is much less likely for developers to overlook unexpected state modifications. For example, Vyper only allows state variables to be modified via syntax such as `self.xyz = 5`. Contrast this to solidity, where a statement such as `xyz = 5;` does not make immediately clear whether it is modifying a local variable or a state variable of the contract.

However, in both languages, the syntax of a function call does not explicitly distinguish whether or not state modifications can happen. When reading code, it is impossible to know what side effects might be caused by a function call without fully understanding the function's body. In a sense, this is equivalent to the concept of *information hiding*, which means that a caller should not be reliant on the internals of a function, but instead only depend on its signature. Thus, we can require functions to explicitly state any possible modifications in their signatures.
Although this significantly improves the situation, it doesn't entirely solve the problem – if we see a function call like `self.foo()`, it's still necessary to go and look at the signature of the function to determine its declared side effects. Again, this may allow unintended side effects to go unnoticed. If the developer is also required to explicitly allow the side effects at the *call site*, then it becomes immediately clear at a glance what exactly is happening.

```
pub fn foo(amount: u256) -> u256
  modifies(self.last_amount)
  modifies(self.n)
{
  self.last_amount = amount;

  let count: u256;
  modify(self.n) {
    count = self.increment();
  }

  return count;
}

fn increment() -> u256
  modifies(self.n)
{
  let result = self.n;
```

```
  self.n += 1;
  return result;
}
```

As you can see in the above example, only the indirect side effect caused by the call to `self.increment()` is required to be explicitly declared. In the case of `self.last_amoount = amount`, it is already clear that the state is being modified without any surrounding declarations.

1.3.1 Side Effects in Reentrant Calls

This concept of side effect declaration also pairs very well with the granular reentrancy path declaration detailed earlier. In fact, without such a feature, it would be almost impossible to determine what side effects might occur to the contract state during an external call. However, if the external call must specify exactly which reentrancies are possible, then the compiler can treat the external call as though it were a regular function call, at least concerning side effects. Thus, the call site must be marked with the potential side effects that would be induced if the reentrant call is made.
Note that we also require the possible reentrancy to be declared in the function signature. If the reentrant call itself may trigger further such calls, it should be clear at the call site that this is a possibility. However, it is unlikely for a real-world use case to strictly require nested reentrant calls, so it would also suffice to simply disallow further reentrant calls.

Listing 1.1: Ours

```
pub fn foo(
  x: u256
) -> u256
  modifies(self.val)
  reenters(Self::bar)
{
  self.val = x;
  modify(self.val) {
    EVM::reentrant_call(
      Self::bar,
      address,
      data
    );
  }
  return self.val;
}

pub fn bar()
  modifies(self.val)
{
   self.val += 1;
}
```

Listing 1.2: Solidity

```
function foo(uint x)
  public
  returns (uint)
{
    value = newValue;
    address.call(data);
    return value;
}

function bar()
  public
{
    value += 1;
}
```

Here, it is again immediately obvious that the new value written to the state variable may be overwritten by the reentrant call. Contrast that to the equivalent Solidity code, where it is not even clear that a reentrant call is possible in the first place, let alone what side effects it may cause. In fact, any of the existing languages for EVM would be equivalent, since their reentrancy protections are declared at the entrypoints, not the call sites.
Of course, this example also makes clear that the strict requirements for side effect declarations result in verbose code. However, we believe that this is a trade-off which is worth the additional overhead, since the stakes are so high.

Chapter 2

Introduction

The median of a dataset is a fundamental measure of central tendency that arises in numerous computational and statistical contexts. In an array of n elements, indexed from 1 to n, the median can be intuitively understood as the "middle" element in the sorted order if n is odd, or the average of the two middle elements if n is even. While this definition naturally suggests sorting the entire array, doing so in $\mathcal{O}(n \log(n))$ time is often unnecessary when the goal is to identify only the median.

Motivation and Previous Work

Several well-known algorithms exist for selecting the median or, more generally, the k-th smallest element of an array. A straightforward method is to sort the array and then pick the median in constant time. However, this approach has a runtime of $\mathcal{O}(n \log(n))$, which can be excessive for large n. More advanced methods, such as the *median-of-medians* algorithm introduced by Blum, Floyd, Pratt, Rivest, and Tarjan [1], achieve a worst-case runtime of $\mathcal{O}(n)$ using a deterministic divide-and-conquer strategy. While optimal in the worst case, the median-of-medians algorithm involves significant overhead, making it less practical in many scenarios. Alternatively, randomized algorithms like Quickselect often achieve linear-time performance on average but can degrade to $\mathcal{O}(n^2)$ in the worst case without additional precautions.

Our Contribution

In this thesis, we present a *randomized* algorithm that computes the median in $\mathcal{O}(n)$ time *with high probability*, thereby combining practical efficiency with strong theoretical guarantees. The key idea is to leverage a carefully chosen random sample of the input array:

1. We extract a subset of elements (the "sample") whose size grows sublinearly with n.
2. Using this sample, we identify pivot elements that, with high probability, closely approximate the true median's position.
3. We then reduce the problem to sorting only a small subset of the original array, thereby achieving near-linear time complexity.

Central to our analysis is the use of **Chebyshev's inequality**. While often introduced in purely statistical contexts, Chebyshev's inequality provides a useful probabilistic bound for ensuring that the sampled elements approximate the median accurately. In essence, it tells us that $\mathbb{P}[|X - \mathbb{E}[X]| \geq k]$ can be kept small if $\mathrm{Var}[X]$ is finite, thereby giving us high confidence that our pivot-based "zoom-in" on the median is correct.

More concretely, we prove the following theorem with our algorithm.

Theorem 2.1 *There exists a randomized algorithm to compute the median that runs in time $\mathcal{O}(n)$ and succeeds with probability $1 - \mathcal{O}\left(n^{-1/4}\right)$.*

Thesis Outline

This thesis is structured as follows:

- **Chapter 3 (Notation and Preliminaries):** We define our notational conventions and offer a brief refresher on Chebyshev's inequality, setting the stage for the randomized analysis to come.
- **Chapter 4 (Algorithm and Analysis):** We detail the randomized median-finding algorithm, present its pseudocode, and provide a rigorous proof of correctness. We also analyze the runtime and show why it remains $\mathcal{O}(n)$ with high probability.
- **Chapter 5 (Conclusion):** We summarize our findings, discuss the algorithm's theoretical and practical implications, and highlight possible directions for future research.

Chapter 3

Notation and Preliminaries

Definitions and Notation

Arrays and Median

Throughout this thesis, we denote the input array of size n by

$$a = (a[1], a[2], \ldots, a[n]).$$

We may also use $a[1 \ldots n]$ to denote an array a of length n. Our goal is to find the *median* of a. Formally, we define:

- If n is odd, the median is the element in the $\lceil n/2 \rceil$-th position of the array when sorted in non-decreasing order.
- If n is even, the median is the element in the $\lceil n/2 \rceil$-th position of the sorted array.

Random Variables and Expectation

In the analysis of our randomized median-finding algorithm, we shall frequently refer to *random variables*. A random variable X is a function from the sample space Ω (representing all possible outcomes of a probabilistic experiment) to the real numbers $\mathbb{R}$.

Expectation. The *expected value*, or *mean*, of X is defined by

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\, \mathbb{P}[\omega],$$

in the discrete case. If X has a probability density function f_X (in the continuous case), then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx.$$

Intuitively, $\mathbb{E}[X]$ indicates the "average" or "central" value of the random variable over many independent trials.

Variance and Standard Deviation

To measure how much a random variable deviates from its mean, we use its *variance*, denoted $\mathrm{Var}[X]$. This is defined as:

$$\mathrm{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

The *standard deviation* of X, denoted σ_X, is the square root of the variance:

$$\sigma_X = \sqrt{\mathrm{Var}[X]}.$$

A smaller variance (or standard deviation) means the random variable's values tend to lie closer to its mean, whereas a larger variance indicates a broader spread of values.

Asymptotic Notation

In describing the time complexity of algorithms, we use $\mathcal{O}(\ldots)$ notation in a standard way:

$$f(n) = \mathcal{O}(g(n)) \quad \Longleftrightarrow \quad \lim_{n\to\infty} \frac{f(n)}{g(n)} < \infty.$$

We may also use $\Theta(\ldots)$, $\Omega(\ldots)$, and other asymptotic notations where necessary.

Chebyshev's Inequality

Chebyshev's inequality is a cornerstone result in probability theory, providing a quantitative bound on how a random variable deviates from its expected value. Formally, let X be any random variable with finite mean $\mathbb{E}[X]$ and finite variance $\mathrm{Var}[X]$. Then, for any $k > 0$,

$$\mathbb{P}[|X - \mathbb{E}[X]| \geq k] \;\leq\; \frac{\mathrm{Var}[X]}{k^2}.$$

Intuition and Relation to Markov's Inequality

To build intuition, recall the simpler *Markov's inequality*, which states that for any nonnegative random variable Y and $t > 0$,

$$\mathbb{P}[Y \geq t] \leq \frac{\mathbb{E}[Y]}{t}.$$

Chebyshev's inequality is often viewed as an application of Markov's inequality to the random variable $(X - \mathbb{E}[X])^2$. Indeed,

$$\mathbb{P}[(X - \mathbb{E}[X])^2 \geq k^2] \leq \frac{\mathbb{E}[\left(X - \mathbb{E}[X]\right)^2]}{k^2} = \frac{\mathrm{Var}[X]}{k^2},$$

and from $\{\, |X - \mathbb{E}[X]| \geq k \,\} \equiv \{\, (X - \mathbb{E}[X])^2 \geq k^2 \,\}$, we obtain Chebyshev's inequality.

The key takeaway is that Chebyshev's inequality captures a broad class of distributions. It requires no assumption of normality or other specific distributional shapes, only finite variance. In exchange for this generality, Chebyshev's bound can be somewhat loose compared to more specialized inequalities (e.g., Hoeffding's, Chernoff's, or Bernstein's). Nonetheless, it remains a powerful and classic tool for bounding "tail events," i.e. events in which X significantly deviates from its mean.

Why We Use Chebyshev's Inequality Here

In our randomized median-finding algorithm, we draw random samples from the input array and use them to approximate the true median. The "rank" of our chosen pivots in the sorted order of the entire array are themselves random variables. We want to show that, with high probability, these rank containt the $\lceil n/2 \rceil$-th position.

By applying Chebyshev's inequality, we obtain a bound on $\mathbb{P}[|X - \mathbb{E}[X]| \geq k]$ for a well-chosen k, where X is the rank of our pivot. In other words, Chebyshev's inequality formally establishes the reliability of our sampling strategy by demonstrating that, except with small probability, our pivots partition the array around the actual median. This allows us to restrict further computation to a subset of the array whose size is guaranteed to be sufficiently small, thereby ensuring linear running time.

Extensions and Alternatives

While Chebyshev's inequality is sufficient for our purposes, more refined analyses might employ other concentration bounds such as Hoeffding's or Chernoff's inequalities. These stronger bounds can yield tighter probabilities when the underlying sample distribution has additional regularity properties (e.g., bounded random variables or independent Bernoulli trials). However, Chebyshev's inequality has the advantage of simplicity and broad applicability, making it an ideal choice for illustrating the core ideas in a clear manner.

Chapter 4

Algorithm and Analysis

In this chapter, we present our main randomized median-finding algorithm in full detail and analyze its performance. We first give a high-level description and provide the pseudocode in Algorithm 1. We then prove two central theorems: one that establishes the algorithm's probability of success (i.e. returning the correct median) and another that bounds its running time by $\mathcal{O}(n)$. Throughout this chapter, we use the notation introduced in Chapter 3.

Algorithm Description

The guiding idea behind the algorithm is to select a "small" random sample from the input array, sort this sample, and use carefully chosen elements from this sample as *pivots* to bracket the true median. Specifically:

- We draw $m = \lceil n^{3/4} \rceil$ elements uniformly at random from the input array $a[1 \dots n]$ and sort these sampled elements into an array $b[1 \dots m]$.
- We then pick two elements from b, denoted p_ℓ and p_r, near the middle of b but offset by $\pm\sqrt{n}$. Intuitively, these two pivots should "sandwich" the true median of the entire array with high probability.
- We gather all elements of the original array a that lie between these two pivots (inclusive) into a temporary array T. We verify that the median must lie in this sub-array, and that T is not excessively large.
- If those checks are satisfied, we sort T and extract the median; otherwise, the algorithm reports a *failure*.

The rest of this chapter gives a proof of two properties of this algorithm:

1. With high probability, Algorithm 1 does not fail and returns the correct median.
2. Conditioned on success, the overall running time is $\mathcal{O}(n)$.

Algorithm 1 RandMedian: Randomized Median Algorithm

Require: Unsorted array $a[1 \dots n]$ of size n.
Ensure: Median of a or *failure*.

Initialize an empty array b.
For each element $x \in a[1 \dots n]$:
  Include x in b **with probability** $n^{-1/4}$, independently of other elements.
$m \leftarrow |b|$ ▷ (Random) size of the sample; note that $\mathbb{E}[m] = n^{3/4}$.
if $m > n^{3/4} + \sqrt{n}$ **then**
  return *failure*
end if
Sort b in $\mathcal{O}(m \log m)$ time
Compute pivots:

$$p_\ell \leftarrow b\left[\left\lfloor \tfrac{m}{2} - \sqrt{n} \right\rfloor\right], \quad p_r \leftarrow b\left[\left\lceil \tfrac{m}{2} + \sqrt{n} \right\rceil\right]$$

Define the subset $T \leftarrow \{\, x \in a : p_\ell \leq x \leq p_r \}$
Compute:

$$s \leftarrow \#\{\, x \in a : x < p_\ell \},$$
$$t \leftarrow \#\{\, x \in a : p_\ell \leq x \leq p_r \}.$$

if $s < \lceil n/2 \rceil$ **and** $s + t \geq \lceil n/2 \rceil$ **and** $t < 4n^{3/4}$ **then**
  Sort T in $\mathcal{O}(|T| \log |T|)$ time
  Median of $a \leftarrow T[\lceil n/2 \rceil - s]$
  return Median of a
else
  return *failure*
end if

Probability of Success

We begin by showing that, except with small probability, the two pivots p_ℓ and p_r are chosen so that the sub-array T contains the true median of a and is not excessively large. Formally, we have:

Theorem 4.1 (Correctness with High Probability) *The probability that Algorithm 1 returns the correct median is at least* $1 - \mathcal{O}\left(n^{-1/4}\right)$. *In other words, the algorithm succeeds* with high probability.

Proof For simplicity, we assume throughout this proof that n is even and that $n^{3/4}$ is an integer; this avoids cumbersome floor and ceiling functions. The general case follows by analogous, but slightly more tedious, arguments.

Recall that $a[1 \dots n]$ is our input array of n distinct elements. Let $b[1 \dots m]$ be the sample of size m drawn by sampling each element with probability

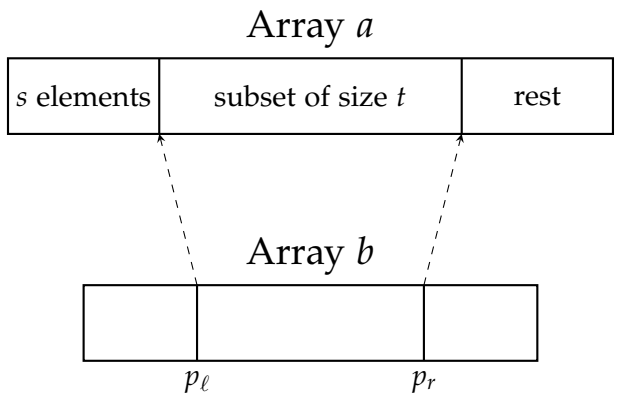

Figure 4.1: Illustration of the arrays and the pivots, showing array a, sampled array b, and the pivots p_ℓ, p_r. Note that this visualization shows the sorted version of the arrays.

$n^{-1/4}$ from a. The setup is also visualized in Figure 4.1. We then sort b, pick the pivots[1]

$$p_\ell \;=\; b\Big[\tfrac{m}{2} - \sqrt{n}\Big], \quad p_r \;=\; b\Big[\tfrac{m}{2} + \sqrt{n}\Big],$$

and collect into T all elements $x \in a$ with $p_\ell \le x \le p_r$. Let

$$s \;=\; \#\{\, x \in a : x < p_\ell \,\} \quad \text{and} \quad t \;=\; \#\{\, x \in a : p_\ell \le x \le p_r \,\}.$$

Key sets and random variables. For any integer $k \in \{0, \dots, n\}$, define

$$S_k \;=\; \{\text{the } k \text{ smallest elements of } a\}.$$

We then define the random variable

$$X_k \;=\; \#\{\, x \in b : x \in S_k \}.$$

Since each element of a is independently included in b with probability $n^{-1/4}$, the number of elements of S_k that appear in b follows

$$X_k \;\sim\; \mathrm{Bin}\big(k,\; n^{-1/4}\big).$$

Hence,

$$\mathbb{E}[X_k] \;=\; k\,n^{-1/4}, \quad \mathrm{Var}[X_k] \;=\; k\,n^{-1/4}\big(1 - n^{-1/4}\big) \;\le\; k\,n^{-1/4}.$$

Application of Chebyshev's Inequality. Since $X_k \sim \mathrm{Bin}(k,\, n^{-1/4})$ with $\mathrm{Var}[X_k] \le k\,n^{-1/4}$, Chebyshev's inequality gives

$$\mathbb{P}\big[\big|X_k - k\,n^{-1/4}\big| \;\ge\; \sqrt{n}\big] \;\le\; \frac{\mathrm{Var}[X_k]}{n} \;\le\; \frac{k\,n^{-1/4}}{n} \;=\; \frac{k}{n^{5/4}}. \tag{4.1}$$

[1]Technically, if m is too small, the following lines are undefined (out of bounds). But this is avoided whp, as for the other events later in the proof.

Defining the "bad" events. We show that the only way the algorithm can fail (i.e. not return the correct median) is if one of the following four events occurs. Denote

$$\begin{aligned} A &:= \{ s \geq n/2 \}, \\ B &:= \{ s + t < n/2 \}, \\ C &:= \{ s < \tfrac{n}{2} - 2\, n^{3/4} \}, \\ D &:= \{ s + t \geq \tfrac{n}{2} + 2\, n^{3/4} \}. \\ E &:= \{ m > n^{3/4} + \sqrt{n} \}. \end{aligned}$$

Notice that E not occuring means that the first "return failure" line is not executed. If none of these events happens, then we must have

$$s < \tfrac{n}{2}, \quad s + t \geq \tfrac{n}{2}, \quad s \geq \tfrac{n}{2} - 2\, n^{3/4}, \quad \text{and} \quad s + t < \tfrac{n}{2} + 2\, n^{3/4}.$$

One can check that these conditions imply exactly

$$\left(s < \tfrac{n}{2}\right) \quad \text{and} \quad \left(s + t \geq \tfrac{n}{2}\right) \quad \text{and} \quad t < 4\, n^{3/4},$$

so the algorithm's final check (line 11 of the pseudocode) succeeds. In that case, our pivot-based bracket $[\, p_\ell,\, p_r \,]$ indeed contains the median, and the algorithm returns the correct value.

Probability of each bad event. We next prove that $\mathbb{P}[A], \mathbb{P}[B], \mathbb{P}[C], \mathbb{P}[D], \mathbb{P}[E]$ are each at most $n^{-1/4}$. Observe:

- Event A means $s \geq n/2$. Hence among the $n/2$ smallest elements of a (i.e. the set $S_{n/2}$), at most $\frac{1}{2}\, n^{3/4} - \sqrt{n}$ of them are drawn into b. In other words,
$$X_{n/2} \;\leq\; \tfrac{1}{2}\, n^{3/4} \;-\; \sqrt{n}.$$
But from $\mathbb{E}[X_{n/2}] = (n/2)\, n^{-1/4} = \frac{1}{2}\, n^{3/4}$ and (4.1), we get $\mathbb{P}[X_{n/2} \leq \frac{1}{2}\, n^{3/4} - \sqrt{n}] \leq n^{-1/4}$. Thus $\mathbb{P}[A] \leq n^{-1/4}$.
- Event B means $s + t < n/2$. Equivalently, the set $S_{n/2}$ has more than $\frac{1}{2}\, n^{3/4} + \sqrt{n}$ elements contained in b. Hence
$$X_{n/2} \;\geq\; \tfrac{1}{2}\, n^{3/4} \;+\; \sqrt{n}.$$
Again, from $\mathbb{E}[X_{n/2}] = \frac{1}{2}\, n^{3/4}$ and (4.1), we get $\mathbb{P}[X_{n/2} \geq \frac{1}{2}\, n^{3/4} + \sqrt{n}] \leq n^{-1/4}$. So $\mathbb{P}[B] \leq n^{-1/4}$.
- Event C means $s < n/2 - 2\, n^{3/4}$. In this case, the set $S_{n/2 - 2\, n^{3/4}}$ must have at least $\frac{1}{2}\, n^{3/4} - \sqrt{n}$ of its elements in b. Formally,
$$X_{n/2 - 2\, n^{3/4}} \;\geq\; \tfrac{1}{2}\, n^{3/4} - \sqrt{n}.$$
Using $\mathbb{E}[X_{n/2 - 2\, n^{3/4}}] = (n/2 - 2\, n^{3/4})\, n^{-1/4} = \frac{1}{2} n^{3/4} - 2\sqrt{n}$ and (4.1) again shows $\mathbb{P}[C] \leq n^{-1/4}$.

- Event D means $s + t \geq n/2 + 2\,n^{3/4}$. Hence the set $S_{\,n/2+2\,n^{3/4}}$ contains at most $\frac{1}{2}\,n^{3/4} + \sqrt{n}$ elements from b. That is,

$$X_{\,n/2+2\,n^{3/4}} \;\leq\; \tfrac{1}{2}\,n^{3/4} \;+\; \sqrt{n}.$$

  A final application of (4.1) yields $\mathbb{P}[D] \leq n^{-1/4}$.

- A similar calculation involving X_n shows that $\mathbb{P}[E] \leq n^{-1/4}$.

Conclusion. By the union bound,

$$\mathbb{P}[A \cup B \cup C \cup D \cup E] \;\leq\; \mathbb{P}[A] + \mathbb{P}[B] + \mathbb{P}[C] + \mathbb{P}[D] + \mathbb{P}[E] \;\leq\; 5 \times n^{-1/4}$$

In other words, with probability at least $1 - 5n^{-1/4}$, *none* of these events occurs. When all five do not occur, the algorithm's checks

$$s < \lceil n/2 \rceil, \quad s + t \geq \lceil n/2 \rceil, \quad t < 4\,n^{3/4}$$

are satisfied, so the algorithm terminates successfully with the correct median. Hence the overall success probability is at least $1 - 5n^{-1/4}$, completing the proof. □

Runtime Analysis

Next, we establish that Algorithm 1 achieves linear time complexity *with high probability*. Note that if the algorithm returns *failure*, it may be retried or combined with a fallback procedure. Here, we focus on the typical execution that succeeds.

Theorem 4.2 (Runtime) *Conditioned on the event that Algorithm 1 does not fail (an event which, by Theorem 4.1, holds with high probability), the total running time is $\mathcal{O}(n)$.*

Proof We break the cost into four parts:

1. **Sampling and Sorting b.** We pick $m \leq n^{3/4} + \sqrt{n} \leq 2n^{3/4}$ elements randomly from a (conditioned on success). This is done in $\mathcal{O}(n)$ time. Sorting b of size m then takes

$$\mathcal{O}(m \log(m)) = \mathcal{O}\Big(n^{3/4} \log\Big(n^{3/4}\Big)\Big).$$

   Since $n^{3/4} \log\big(n^{3/4}\big)$ is asymptotically smaller than n for large n, this component is $o(n)$.

2. **Computing Ranks and Sub-array T.** We determine how many elements of a lie below p_ℓ, how many lie above p_r, etc. This amounts to a single pass through a, comparing each element of a to p_ℓ and p_r. Hence, it costs $\mathcal{O}(n)$ time.

3. **Sorting T.** By Algorithm 1, we only proceed to sort T if $t < 4n^{3/4}$. Thus,
$$t \leq 4n^{3/4}.$$
Sorting T via a standard comparison-based algorithm costs
$$\mathcal{O}(|T|\log(|T)|) \leq \mathcal{O}\Big(n^{3/4}\log\Big(\big(n^{3/4}\big)\Big)\Big),$$
which again is $o(n)$.

4. **Extracting the Median.** After sorting T, we simply pick the element of T whose rank corresponds to $\lceil n/2 \rceil - s$. This is clearly an $\mathcal{O}(1)$ operation.

Summing these components, the *dominant term* is $\mathcal{O}(n)$ (from the pass to identify s and t). The sublinear sorting costs on b and on T both constitute $\mathcal{O}\big(n^{3/4}\log(n)\big)$, which is strictly smaller than n for large n. Consequently, the total running time is $\mathcal{O}(n)$. □

In conclusion, the previous two theorems immediately prove Theorem 2.1, our main result, which we restate here for convenience.

Theorem 2.1 *There exists a randomized algorithm to compute the median that runs in time $\mathcal{O}(n)$ and succeeds with probability $1 - \mathcal{O}\big(n^{-1/4}\big)$.*

Discussion and Extensions

Theorems 4.1 and 4.2 jointly show that Algorithm 1 achieves $\mathcal{O}(n)$ time with high probability of success. In rare cases where the check fails, the algorithm returns *failure* rather than risking an incorrect answer. One can either re-run the algorithm (the probability of repeated failures being very low) or combine it with a fallback deterministic linear-time median-finding routine such as the median-of-medians approach. The randomized algorithm typically attains lower hidden constants in practice than purely deterministic linear-time solutions, thereby offering an attractive balance of theoretical guarantees and real-world performance.

In the next chapter, we summarize our results and highlight possible directions for future work on randomized selection algorithms.

Chapter 5

Conclusion

In this thesis, we presented a randomized selection algorithm for identifying the median of an unsorted array in expected linear time. Our method hinges on sampling a sublinear number of elements, sorting them to obtain pivots, and then "zooming in" on a small region of the array that is guaranteed, with high probability, to contain the true median. The crux of our analysis relied on Chebyshev's inequality, which provided a principled way to bound the probability that our random pivots deviate significantly from the actual median's rank in the sorted order.

Summary of Contributions.

- We introduced a randomized median-finding algorithm that leverages a sample of size roughly $n^{3/4}$. Through sorting this sample and selecting pivots near its middle, the algorithm narrows down the search space to a sub-array whose size is no more than $4n^{3/4}$.
- We proved a high-probability bound (Theorem 4.1) on the correctness of the algorithm. Specifically, with probability at least $1 - \mathcal{O}(n^{-1/4})$, our pivots capture the true median, ensuring that the algorithm does not report *failure*.
- We established a near-linear worst-case running time (Theorem 4.2). The cost of sorting the small sample and, conditionally, the sub-array T both remain asymptotically lower than $\mathcal{O}(n)$, leaving the dominant cost to be a single pass over the array to compute element ranks relative to the pivots. Hence, the overall algorithm runs in $\mathcal{O}(n)$ time with high probability.
- Beyond the immediate theoretical guarantees, we discussed how to combine our randomized approach with a deterministic "median-of-medians" fallback to handle rare failure events, ensuring correctness

remains absolute. This hybrid strategy preserves the practical advantages of randomization while delivering a worst-case linear-time bound.

Limitations and Possible Extensions. While our use of Chebyshev's inequality is conceptually simple and broadly applicable, tighter concentration results (e.g. Chernoff or Hoeffding bounds) could potentially yield smaller failure probabilities or allow an even smaller sample size. However, such refinements might require more specific assumptions about the distribution of the input data or the independence structure of the sampling process.

Another direction for future exploration is to extend the approach to streaming and dynamic settings. In scenarios where the dataset arrives incrementally, maintaining a small, representative sample to approximate the median could be achieved through reservoir sampling or related techniques. Additionally, one might investigate parallel or distributed variants of the algorithm, where the sampling and pivot-selection steps are split across multiple processors or machines.

Bibliography

[1] Manuel Blum, Robert W Floyd, Vaughan R Pratt, Ronald L Rivest, and Robert E Tarjan. Time bounds for selection. *Journal of Computer and System Sciences*, 7(4):448–461, 1973.

Dummy Appendix

You can defer lengthy calculations that would otherwise only interrupt the flow of your thesis to an appendix.

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